Abstract

Educational campuses handle a large number of day-to-day issues related to Wi-Fi, electricity, laboratories, library, hostels, transport, security and furniture. These issues are usually reported through phone calls, paper applications or word of mouth, which leaves no record, no accountability and no way for the complainant to know the status of a request. As a result, complaints are delayed, duplicated or lost, and the administration has no data with which to improve its services.

This project presents **CampusCare**, **a Smart Campus Issue Management and Resolution System** developed as a **Windows desktop application**. The system provides a single digital channel through which students submit complaints with a category, location, priority, description and supporting attachments. Each complaint is automatically routed to the responsible department, assigned to a staff member by the administrator, and moved through a seven-stage lifecycle: submitted, assigned, accepted, in progress, resolved, closed and reopened. Every status change is stored with remarks, the person responsible and a timestamp, giving a complete audit history. Users receive in-app notifications on each change, students can rate the resolution, and administrators can view analytics by status, priority, department and category, and export reports as CSV files.

The application follows **a layered architecture of controller, service, data-access and model layers**. The user interface is built with **Java 21, JavaFX and FXML**, and the system uses **PostgreSQL** hosted on the cloud, accessed through JDBC, so that students, staff and administrators working on different computers see the same data. Role-based access control separates the student, staff and administrator views, passwords are protected with **BCrypt** hashing, and input validation is applied on all forms. The project is built with Maven and packaged as a standalone Windows executable with jpackage. The team followed a Git-based workflow with separate feature branches for each module.

CampusCare replaces informal complaint handling with a transparent, traceable and measurable process, and its analytics help campus authorities identify recurring problems and improve response quality.

Keywords: campus issue management, complaint tracking, **JavaFX**, **PostgreSQL**, **role-based access**, **desktop application**, workflow automation